# Lecture Notes for Math 243:
# Calculus III

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# 0 Tentative Course Outline

- **Weeks 1-3:** Parametric equations and Polar coordinates
  1. Arc length
  2. Curves defined by parametric equations.
  3. Calculus with plane curves.
  4. Polar coordinates
  5. Areas and lengths in polar coordinates
  6. Conic sections.
- **Weeks 4-6:** Vectors and the geometry of space.
  1. Three-dimensional coordinate systems.
  2. Vectors.
  3. The dot product.
  4. The cross product.
  5. Equations of lines and planes.
  6. Cylinders and quadric surfaces.
- **Friday, October 16 (week 8):** Midterm exam
- **Weeks 7-9:** Vector functions.
  1. Vector functions and space curves
  2. Derivatives and integrals of vector functions.
  3. Arc length and curvature.
  4. Motion in space: velocity and acceleration.
- **Weeks 10-16:** Partial derivatives.
  1. Functions of several variables.
  2. Limits and continuity
  3. Partial derivatives.
  4. Tangent planes and linear approximations
  5. The chain rule.
  6. Directional derivatives and the gradient vector.
  7. Maximum and minimum values.
  8. Discovery project (p.1010 Stewart): Quadratic approximation and critical points (Taylors formula for two variables).
  9. Lagrange multipliers.
- **Monday, December 14:** Final exam.

# 1 2026-08-24 | Week 01 | Lecture 01

*This lecture is based on textbook section 6.3. “Arc length”*

> ***The nexus question of this lecture:*** *How do we compute the length of a curve, assuming the curve is the graph of a function?*

## 1.1 Arc Length

First, recall the definition of ***derivative***:

$$f'(x) := \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

or equivalently,

$$f'(x) = \lim_{\tilde{x}\to x} \frac{f(\tilde{x})-f(x)}{\tilde{x}-x}$$

In other words, the derivative is obtained by computing

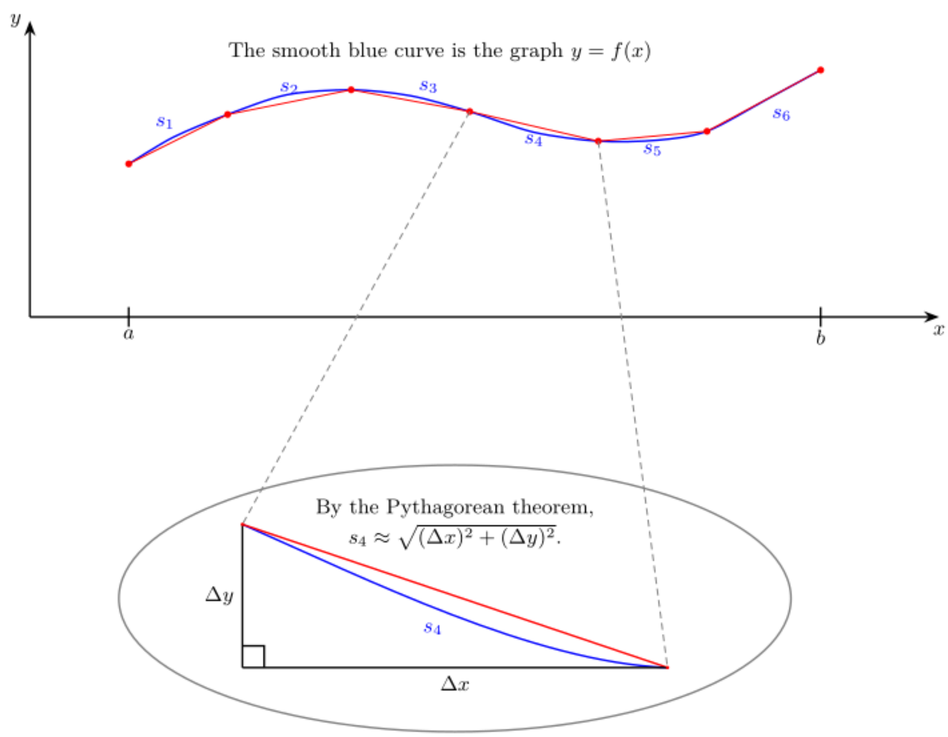
$$\frac{\text{change in } y\text{-value}}{\text{change in } x\text{-value}} = \frac{\Delta y}{\Delta x}$$

as $\Delta x \to 0$.

One of the central ideas in calculus is to approximate curvy things with flat things. For example, if faced with a curve, approximate it with straight lines. For example,

**Problem:** I am an ant, and I want to travel from $a$ to $b$ by walking on the graph of $f(x)$. How far must I walk?

**Strategy:** Cheat a little: think of the ant walking in a bunch of straight lines that are pretty close to the curve.



In this example, the length of the curve $y = f(x)$ from $x = a$ to $x = b$ can be decomposed as

$$\text{(Arc Length)} = s_1 + s_2 + \ldots + s_6,$$

where

$$s_i = \text{(length of segment } i\text{)}$$

Each segment length can be approximated by a straight line. For example,

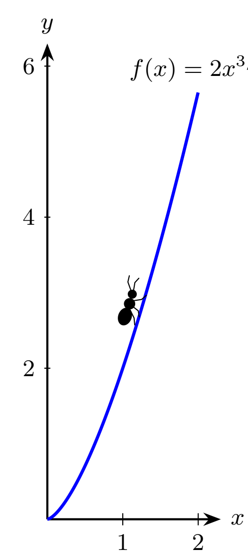
$$\begin{aligned}
s_4 &\approx \sqrt{(\Delta x)^2 + (\Delta y)^2} && \text{since the line segment length approximates the curve length} \\
&= \sqrt{\left[1 + \left(\frac{\Delta y}{\Delta x}\right)^2\right](\Delta x)^2} \\
&= \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}\,\Delta x \\
&\approx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx && \text{this holds when } \Delta x \approx 0 \text{, so long as the curve is smooth} \\
&= \sqrt{1 + [f'(x)]^2}\,dx && \text{by change of notation}
\end{aligned}$$

This example motivates the following theorem.

**Theorem 1** (Arc length formula)**.** *Let $f$ be a differentiable function on the interval $[a, b]$, and assume that $f'$ is continuous on $[a, b]$. Then the arc length of $f$ from $x = a$ to $x = b$ is*

$$\textit{(Arc Length)} = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$$

**Example 2** (Arc length)**.** Find the length of the curve defined by $f(x) = 2x^{3/2}$, where $x$ ranges from $x = 0$ to $x = 2$.




*Solution:* $f'(x) = 3x^{\frac{1}{2}}$, so

$$\begin{aligned}
\text{(Arc length)} &= \int_0^2 \sqrt{1 + [f'(x)]^2}\,dx \\
&= \int_0^2 \sqrt{1 + 9x}\,dx
\end{aligned}$$

which we can solve with a $u$-substitution, taking $u = 1 + 9x$. Doing this gives a length of about 6.06.

End of Example 2. $\square$

The ***hyperbolic cosine*** and ***hyperbolic sine*** functions are defined as

$$\cosh(x) := \frac{e^x + e^{-x}}{2}, \qquad \sinh(x) := \frac{e^x - e^{-x}}{2}.$$

These two functions are "nice" in that they behave in some respects similarly to the standard sine and cosine functions; in particular, cosh is an even function (like $\cos(x)$) and $\sinh(x)$ is odd (like $\sin(x)$).

One can check they satisfy:

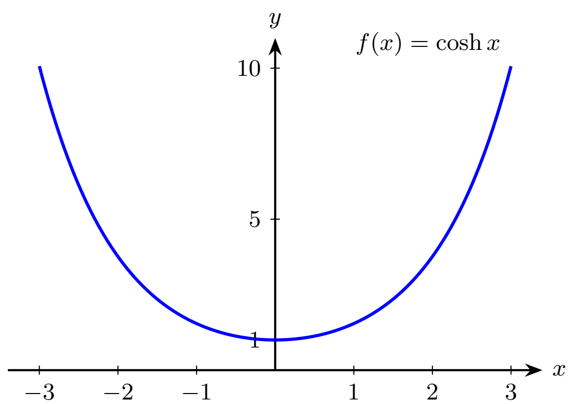
$$\cosh'(x) = \sinh(x), \qquad \sinh'(x) = \cosh(x). \tag{1}$$

Also, we have the following identity:

$$\cosh^2(x) - \sinh^2(x) = 1. \tag{2}$$

A hanging chain forms a curve known as a ***catenary***. The general formula of a catenary is a function involving hyperbolic cosine; in the next example, we'll consider the simplest case.

**Example 3** (Arc length)**.** Suppose $f(x) = \cosh(x)$ represents a chain from two endpoints at $x = -3$ and $x = 3$. Find the length of the chain.




$$\begin{aligned}
(\text{length of chain}) &= \int_{-3}^{3} \sqrt{1 + \left[\cosh'(x)\right]^2} dx \\
&= \int_{-3}^{3} \sqrt{1 + \sinh^2(x)} dx && \text{since } \cosh'(x) = \sinh(x) \text{ by Eq. (1)} \\
&= \int_{-3}^{3} \sqrt{\cosh^2(x)} dx && \text{by Eq. (2)} \\
&= \int_{-3}^{3} \cosh(x) dx && \text{since } \cosh(x) > 0 \text{ for all } x \in \mathbb{R} \\
&= [\sinh(x)]_{x=-3}^{x=3} && \text{since } \sinh'(x) = \cosh(x) \text{ Eq. (1)} \\
&= \sinh(3) - \sinh(3) \\
&\approx 20.036
\end{aligned}$$

End of Example 3. $\square$

**Tips**

- Check that your answer is positive. Arc length is never negative.
- These problems are usually contrived so that you can actually do the integral. If the integral looks took hard, double check your work. Also, there might be a trick to the integral.